



Advanced Computer Vision Final Presentation

Coin Classifier

Presented by

Abinand Balajee (47976586)

Shasanka Acharya (48004707)

Prajwal Chaudhary (47984805)

Sushrut Kuchik (47944250)

Introduction

Motivation

- Coin classification is challenging due to fine visual differences (e.g., texture, embossing, size)
- Variability in lighting and camera quality further complicates recognition

Use Case

- Enable real-time coin recognition on a robotic platform
- Supports autonomous decisions through visual input

Project Objective

- Build a robust model to classify coin denominations across countries
- Deploy the model within **ROS2**, enabling perception-to-action control in the robot.

Introduction

Dataset Overview

- 48 coin classes from 8 countries (USA, India, Nepal, Japan, UK, Canada, NZ, Australia)
- All denominations included
- Images manually sourced for diversity in:
 - Lighting, background, angle, and quality
 - Metallic finish, size, and design details

Phase 1 & 2 Summary

- Trained & evaluated 8 CNN models on the full dataset
- **WideResNet** performed best with **93.7% test accuracy**
- Accuracy improved with:
 - Fine-tuning
 - Data augmentation for real-world variability

Approaches

Data Collection

- 1) **Source:** Google Images, numismatic sites, auction platforms, collector forums
- 2) **Folder structure:**
CoinImages/ → subfolders by country → subfolders by denomination → files named with h or t

Preprocessing Pipeline:

- 1) **Resize** to 224×224 px
- 2) **Grayscale + Median Blur** for noise reduction, preserve coin edges
- 3) **Hough Circle Crop** if coin detected, crop to circle; else keep full image
- 4) **Convert BGR to RGB** (PyTorch convention)
Normalize pixel values to $[0, 1]$
- 5) **Augmentation (on-the-fly):** Random rotation $\pm 5^\circ$, Horizontal flip, brightness/contrast jitter

Model Training Phases & Deployment

- **Phase 1: Baseline (no fine-tuning)**
 - 8 pretrained models (AlexNet, VGG16, RegNet, WideResNet, EfficientNetV2-B0, ResNet-50, ResNet-18, GoogLeNet)
 - Best test accuracy: **ResNet-50** → **88%**
- **Phase 2: Fine-Tuning + Augmentation**
 - Last layer unfrozen, LR scheduling, on-the-fly augment
 - Top result: **WideResNet** → **93.7%** test accuracy

Phase 3 Adaptation

- Reduced to 3 representative coin classes:
 - Japanese 10 yen, Australian 2 AUD, US 1 cent
 - Chosen for visual and material diversity + practical image availability

Workflow in Phase 3



Results

Models	Training Accuracy %	Testing Accuracy %
AlexNet	97.18	73.49
VGG16	99.17	84.75
RegNet	69.79	78.18
WideResNet	90.43	93.70
EfficientNetV2-B0	85.00	82.00
ResNet-50	80.00	69.00
Resnet-18	99.61	89.00
Googlenet	94.23	85.00

Evaluation of Phase 2 and Phase 3

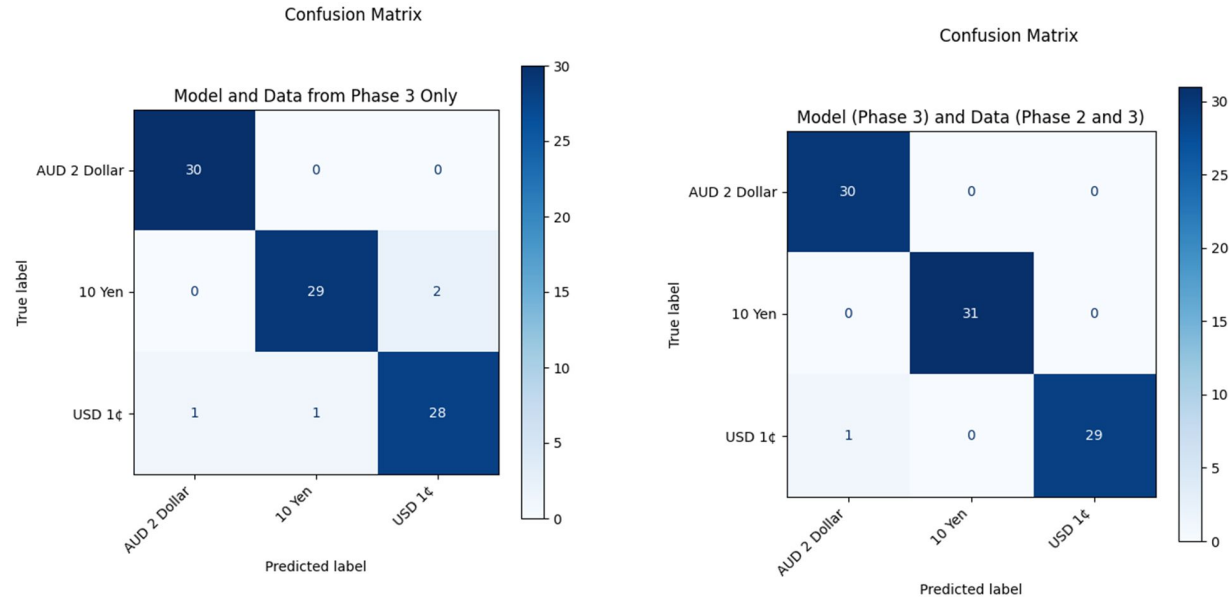


fig: Confusion matrices for - Model and Data from Phase 3 only (left); Model (Phase 3) and Data (Phase 2 and 3) (right)

Discussions and Analysis

What worked for us?

1. Collecting images of high quality and proper labelling.
2. Proper preprocessing techniques
3. Using mild data augmentation techniques
4. Experimenting with different CNN models

What did not work for us?

1. Unable to classify coin images correctly, when clicked directly by HiWonder camera
2. Have a more streamlined flow to move the robot.

Discussions and Analysis

Common patterns of errors

1. Different coins having similar 'heads side' were classified incorrectly. Example: GBP 1 penny and NZ 1 dollar



Improvements

1. Design a more real world application, example vending machine using HiWonder's robot arm.
2. Can work on improving the predictions for the above mentioned problem.

Contributions

Name	Contributions
Abinand Balajee	● Collected and prepared coin image data for Japan and Canada (13 classes) ● Built a custom data loader and trained ResNet-18 and GoogleNet using techniques like augmentation and learning rate scheduling ● Captured images for Phase 3 training (Japanese 10 yen, Australian 2 AUD, US 1 cent) ● Integrated the model with ROS2 to control robot movement and image capture functions
Shasanka Acharya	● Collected coin images for Nepal and New Zealand(10 classes). ● Worked on the data-preprocessing pipeline. ● Worked with ResNet-50 and EfficientNetV2-B0 models and fine tuned them. ● Worked on operating the robot's camera to capture 150 additional images for fine tuning and deploying the model in phase 3. ● Debugging errors during deployment and classification.
Prajwal Chaudhary	● Collected coin images for Australia and United Kingdom (14 classes). ● Conducted data exploration. ● Worked with RegNet and WideResNet models and fine tuned them. ● Worked on moving the robot and figuring out the camera controls.
Sushrut Kuchik	● Collected coin images for United States and India(11 classes). ● Worked with Alexnet and VGG models and fine tuned them. ● Deploying WideResenet model and integrating it with robot's movements. ● Debugging the runtime errors encountered during integration.

THANK YOU!